Weather Trend Forecasting with the Global Weather Repository

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Abstract—This report analyses 170 928 daily weather snapshots for 197 world capitals (May 2024 – October 2026) from the Kaggle Global Weather Repository. It describes a reproducible pipeline (DVC + MLflow) for cleaning, EDA, anomaly detection, forecasting and advanced analyses. Two data issues stand out. The local hour of the daily snapshot drifts from afternoon to early morning, which fakes a cooling of −1.8 °C/yr, and one whole day of data is corrupted at the source. With both handled, a stacked ensemble forecasts next-day temperature with an RMSE of 1.84 °C, 12 % better than persistence, and harmonic regression forecasts the global mean 60 days ahead with an RMSE of about 0.35 °C. The report also covers climate patterns, air quality, feature importance, and spatial and geographic structure.

Keywords— *weather forecasting, time series, EDA, anomaly detection, ensembles, SHAP, spatial analysis, MLflow, DVC*

PM Accelerator — Mission

PM Accelerator (Dr. Nancy Li) is a global community that helps aspiring and current product managers fast-track their careers through coaching, an AI Product Management bootcamp and hands-on product experience. Its non-profit arm, *PMA Kids*, offers free product-management education to teenagers from underserved families:

"Our mission is to break down financial barriers and achieve educational fairness. With the goal of establishing 200 schools worldwide over the next 20 years, we aim to empower more kids for a better future in their life and career, simultaneously fostering a diverse landscape in the tech industry."

Source: https://www.pmaccelerator.io/

1. Introduction

The *Global Weather Repository* (Kaggle) publishes one weather snapshot per day for the capital city of almost every country, with 41 columns covering temperature, wind, pressure, precipitation, humidity, cloud, visibility, UV, six air-quality pollutants, and astronomical data. The version analysed here (v1106) holds **170 928 rows** from **16 May 2024 to 13 October 2026**.

The goal is to forecast future weather trends and to show both basic and advanced data-science practice. The whole project is one reproducible pipeline. **DVC** versions the data and models and chains the stages. **MLflow** tracks every stage, model, metric, and artefact, and holds the model registry. The pipeline has eight stages:

1. **download**: fetch the CSV from Kaggle;
2. **clean**: validity checks, de-duplication, corrupted-batch detection, outlier capping, imputation;
3. **features**: daily panel, lags, rolling statistics, calendar and diurnal features, time split, normalisation;
4. **eda**: distributions, temperature and precipitation trends, correlations, decomposition;
5. **anomaly**: Isolation Forest, LOF, temporal robust z-scores, batch-level detection;
6. **train** / **train_global**: next-day city models and multi-step global-series models;
7. **ensemble**: mean, inverse-RMSE-weighted and stacked ensembles;
8. **analyses**: climate, air quality, feature importance, spatial, and geographic patterns.

2. Data cleaning and preprocessing

Cleaning is in src/weather/data/clean.py. Every count below is written to reports/metrics/clean_report.json and logged to MLflow.

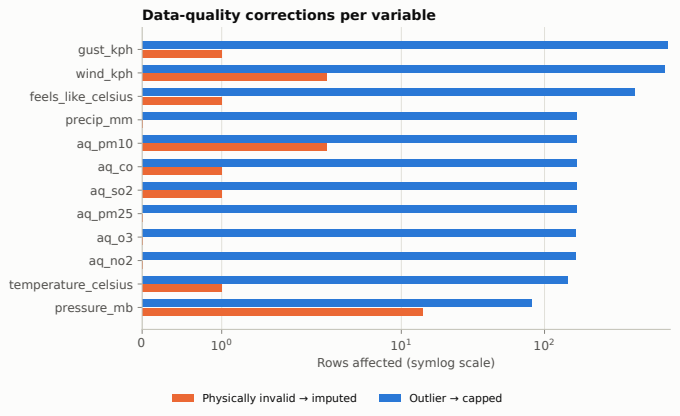


Figure 1. Rows corrected per variable: physically invalid values (imputed) and outliers (capped).

Schema. The seven imperial duplicates (°F, mph, inches, miles) are dropped. The pollutant columns are renamed (`aq_pm25`, …), `last_updated` is parsed to a timestamp, and the snapshot's *local clock hour* is kept as a covariate (Section 3.2).

Coverage and duplicates. 71 of the 268 location strings appear on fewer than 365 days (3 790 rows). They are one-off or mis-named entries and are removed, which leaves **197 capitals**. Countries reported under translated names (e.g. Portuguese *Polônia*, plus Cyrillic and Chinese spellings) are mapped back to the canonical name each location uses most often. Continents come from `pycountry-convert` plus a small override table. 385 same-day duplicate snapshots are collapsed, keeping the latest one. Some locations are *geocoding homonyms* (e.g. "Yaren" in Türkiye, "Bogot" in Bulgaria). Their coordinates match the listed country, so their readings are valid weather for those coordinates and they are kept.

Physically invalid values. Values outside plausible surface ranges become missing. Examples are pressure of 3 006 mb, wind of 2 963 kph, and the sentinel −9999 in CO and SO_2.

Corrupted batches. For each day we take the median, across all cities, of the absolute difference between a city's temperature and its centred 15-day rolling median. A normal day sits near 1 °C. On **17 August 2026** it jumps to 8.5 °C: Riyadh reads 15.6 °C and Bishkek 4 °C in mid-August. That day's weather values (186 rows) are nulled and re-imputed (Figure 12).

Outliers. Roughly symmetric variables (temperature, feels-like, wind, gust, pressure) are capped at a per-city fence $[Q_1 - 3\,\mathrm{IQR},\ Q_3 + 3\,\mathrm{IQR}]$. Heavy-tailed variables (precipitation and the pollutants) are capped at the global 99.9th percentile. Capping (winsorising) keeps the row and stores a boolean flag, so these points can still be studied in Section 4. Visibility is left out of the fence because it is reported as exactly 10 km most of the time (IQR = 0).

Missing values. Within each city, gaps are filled by time interpolation (up to 3 days), then by the city median. Categorical gaps take the city mode. The raw file has no empty cells; every missing value comes from the checks above.

Normalisation. A `StandardScaler` is fitted on the *training window only* and persisted (models/scaler.joblib). It is applied inside the linear model's pipeline. Anomaly detection uses a *city-local* standardisation instead (Section 4).

Table 1. Summary statistics of the cleaned daily panel.

	mean	std	min	median	max
Temperature	21.46	9.26	-29.80	23.60	49.20
Feels like	22.17	11.12	-36.70	24.90	51.20
Humidity	66.80	23.48	2.00	72.00	100.00
Pressure	1014.11	6.78	947.00	1014.00	1052.00
Wind	12.66	8.19	3.60	10.80	91.10
Gust	18.09	11.08	3.60	15.30	119.90
Cloud	40.16	34.82	0.00	29.00	100.00
Precip.	0.13	0.45	0.00	0.00	6.27
UV index	3.07	3.48	0.00	1.40	16.30
Visibility	9.49	2.59	0.00	10.00	32.00
PM2.5	22.49	30.57	0.10	13.14	362.83
PM10	44.52	114.57	0.10	18.70	2040.47
O3	58.05	31.18	0.00	54.00	246.10
NO2	13.99	21.99	0.00	5.46	178.52
CO	412.16	608.27	53.85	264.76	8976.42
SO2	9.64	20.84	0.00	2.30	213.38

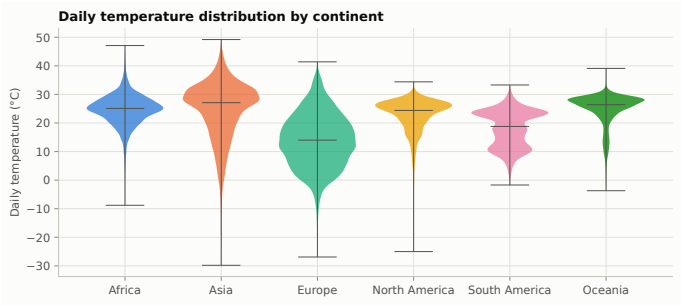


Figure 2. Daily temperature by continent.

Feature engineering. The cleaned data is put on a regular daily grid per city. Features use only information available at day t:

- today's temperature and its lags at 1, 2, 3, 7 and 14 days;
- rolling means and standard deviations over 3, 7 and 14 days, and the day-on-day difference;
- today's humidity, pressure, wind, gust, cloud, precipitation, UV, visibility and feels-like, plus changes in humidity and pressure;
- annual harmonics of the target date, also in a hemisphere-aware version;
- diurnal harmonics of the snapshot hour;
- latitude, longitude, |latitude| and continent.

The target is temperature at $t + 1$. The split is strictly by target date: the test window is the last 60 days (15 Aug–13 Oct 2026, 11 213 rows), validation is the 60 days before it (11 220 rows), and training is everything earlier (142 674 rows).

3. Exploratory data analysis

3.1. Temperature

Figure 2 shows that Africa and Oceania (mostly tropical capitals) are warm with narrow distributions. Europe and Asia span more than 40 °C over the year. The monthly curves (Figure 3) show the two hemispheres in opposite phase, with Europe swinging the most.

3.2. A hidden confounder: the snapshot hour

The global mean temperature of the 184 well-covered capitals (Figure 4) appears to cool sharply. The cause is the *local clock time of the snapshot*. Its median drifts from 16:30 in mid-2024 to 07:45 in late 2026 (Figure 5), so recent observations are increasingly taken near the daily minimum. We regress the series on a linear trend, three annual harmonics and the mean diurnal harmonics (Figure 6). This shrinks

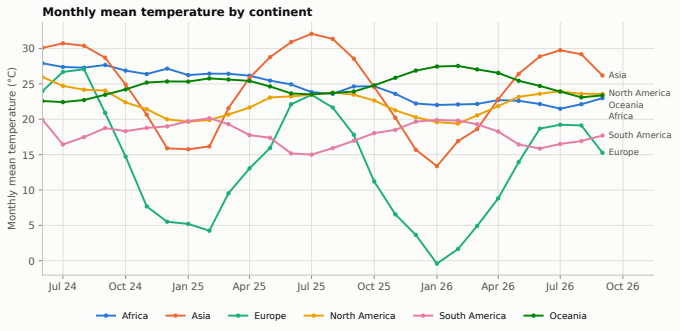


Figure 3. Monthly mean temperature per continent (partial first and last months removed).

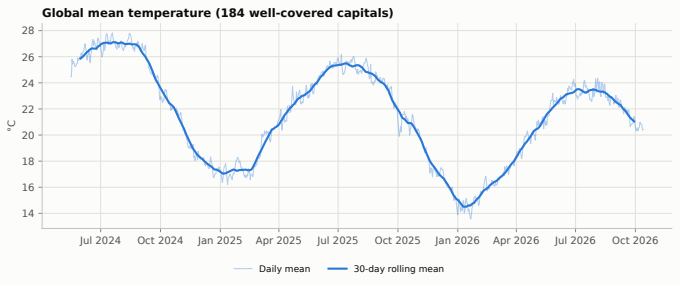


Figure 4. Global mean temperature, daily and 30-day rolling mean.

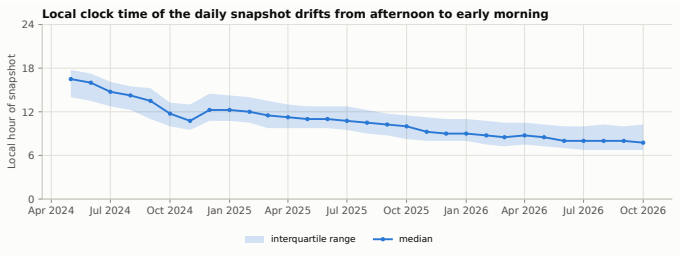


Figure 5. Local hour of the daily snapshot (median and IQR across cities).

the apparent trend from −1.80 to −0.26 **°C/yr**, leaving an annual cycle with an amplitude of about 9.4 °C. Without this correction, any trend model would mistake a change in sampling for climate change.

3.3. Precipitation

Precipitation is zero-inflated: 67 % of snapshots record no rain, and wet-day amounts are roughly log-normal (Figure 7). Seasonality differs by region (Figure 8). Asia and Oceania peak in the boreal-summer monsoon, while South America and Africa follow the migration of the ITCZ.

3.4. Correlations

Spearman correlations (Figure 11) match physical expectations:

- temperature tracks feels-like and UV ($\rho \approx 0.44$);
- humidity is anti-correlated with temperature ($\rho \approx -0.32$) and with visibility;
- PM2.5 and PM10 are almost collinear;
- ozone rises with UV and falls with humidity.

4. Anomaly detection

Raw values would flag every Arctic or desert capital simply for being itself. Anomalies are therefore defined relative to each city's own climate: temperature is standardised against the city-and-month mean, and the other variables against the city mean. Four complementary detectors are used.

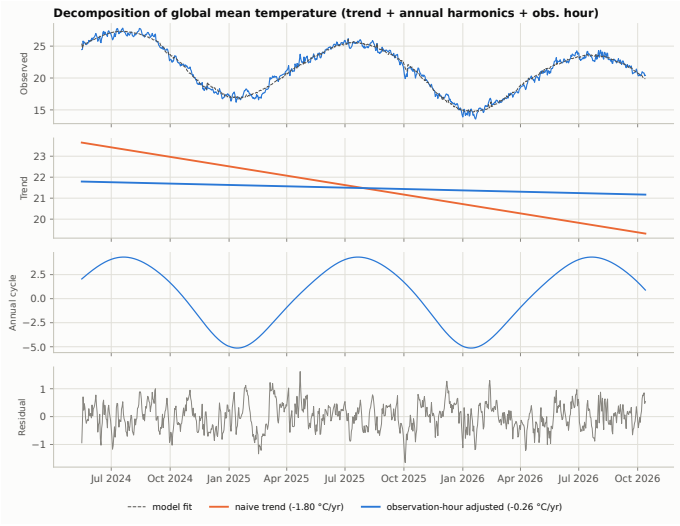


Figure 6. Harmonic-regression decomposition of global mean temperature. STL is degenerate with only about 2.4 annual cycles, so it is not used.

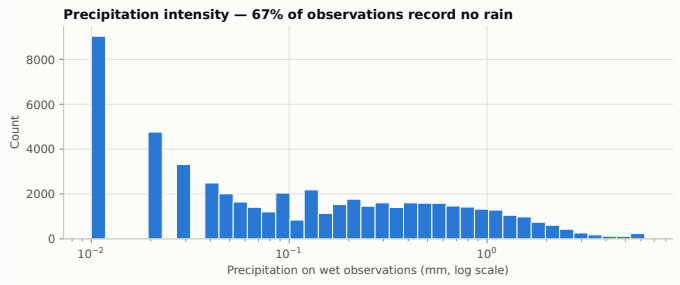


Figure 7. Wet-observation precipitation intensity (log scale).

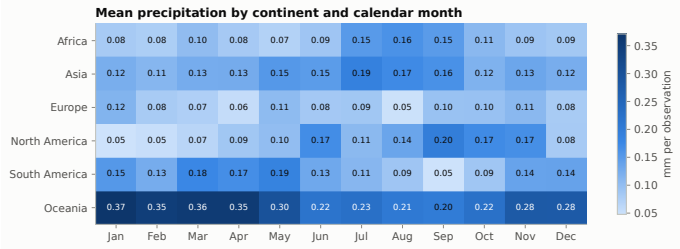


Figure 8. Mean precipitation by continent and calendar month.

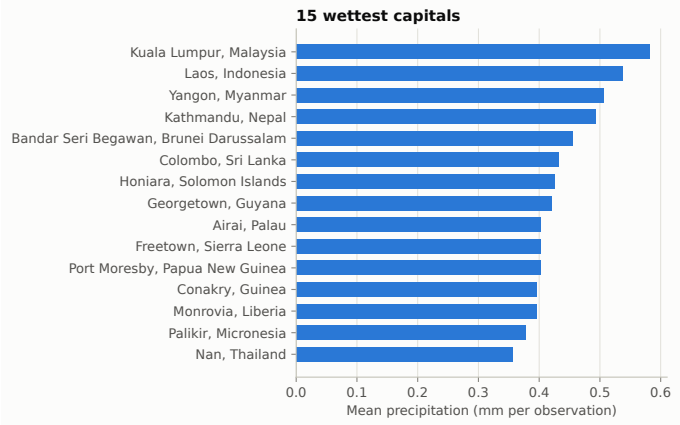


Figure 9. Wettest capitals (left) and most frequent reported conditions (right). (left panel).

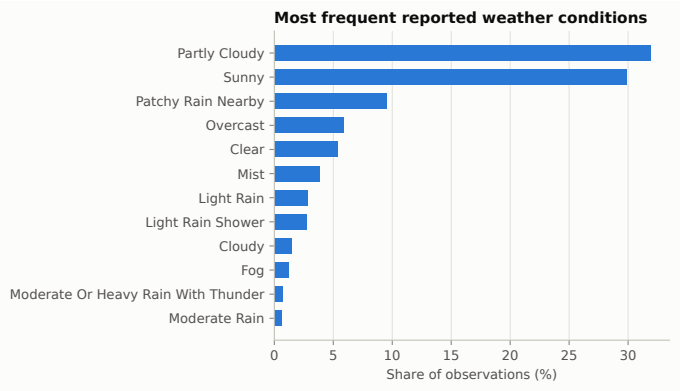


Figure 10. Wettest capitals (left) and most frequent reported conditions (right). (right panel).

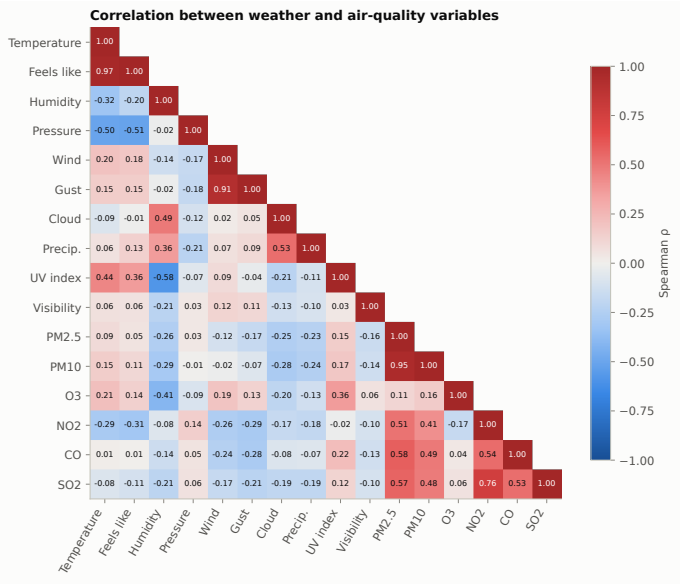


Figure 11. Spearman correlation matrix of weather and air-quality variables.

1. **Isolation Forest** (300 trees, 1 % contamination) on ten locally standardised variables. It flags 1 668 observations.
2. **Local Outlier Factor** ($k = 35$) on a 60 000-row sample.
3. **Temporal robust z-score**: temperature against a centred 15-day rolling median, scaled by the city's MAD, with $|z| > 3.5$. This flags 2.6 % of observations, mostly sudden cold snaps or heat spikes.
4. **Batch-level detector** (Figure 12): days on which most cities deviate at once. This is a data-pipeline failure, not weather.

Isolation Forest anomalies are driven mainly by extreme precipitation, low visibility and humidity (Figure 13): storms and fog events. Isolation Forest and LOF overlap only moderately (Jaccard ≈ 0.11). The global method looks for rare *combinations*, while LOF finds sparse neighbourhoods. About 18 % of Isolation Forest anomalies are also temporal anomalies, and about 19 % contain a value capped during cleaning. The multivariate view therefore adds information beyond univariate fences.

5. Forecasting

Two framings use the `last_updated` time axis.

- **City panel, next-day forecast**: one global model learns from all 197 capitals at once.
- **Global mean series, 60-day multi-step forecast**: the world-wide trend is projected forward.

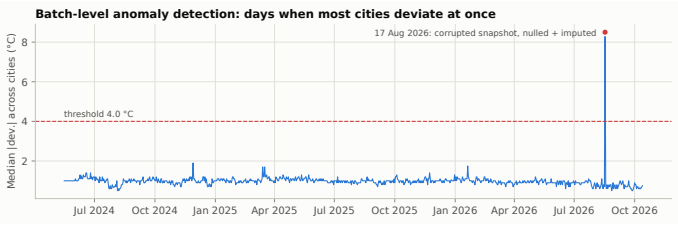


Figure 12. Batch-level detector: the 17 Aug 2026 snapshot is corrupted and was nulled and imputed during cleaning.

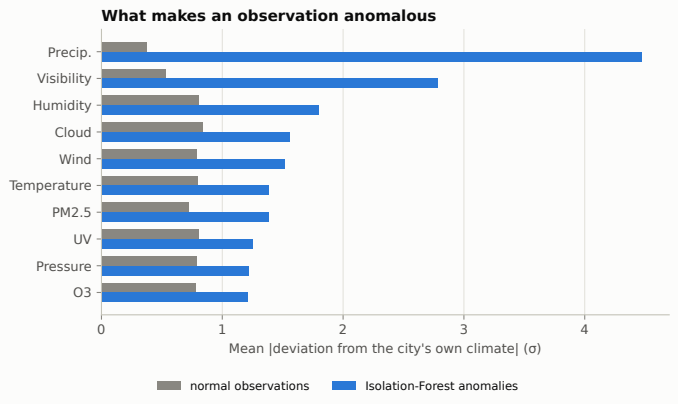


Figure 13. What makes an observation anomalous, and where anomalies sit in the locally standardised weather space. — Drivers.

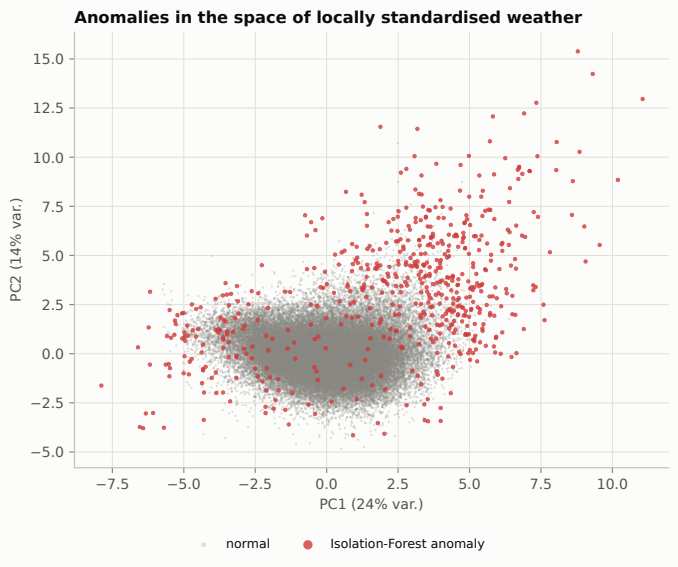


Figure 14. What makes an observation anomalous, and where anomalies sit in the locally standardised weather space. — PCA projection.

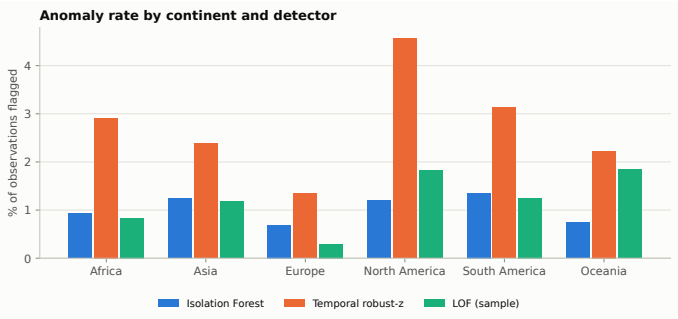


Figure 15. Anomaly rate by continent and detector.

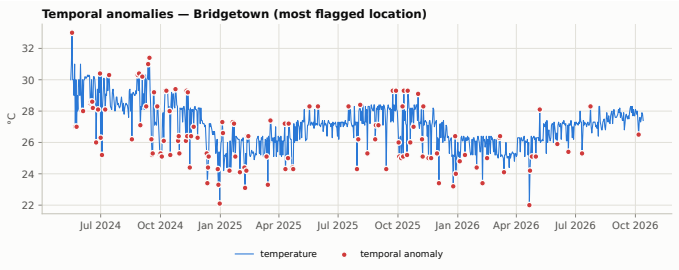


Figure 16. Temporal anomalies for the most-flagged capital.

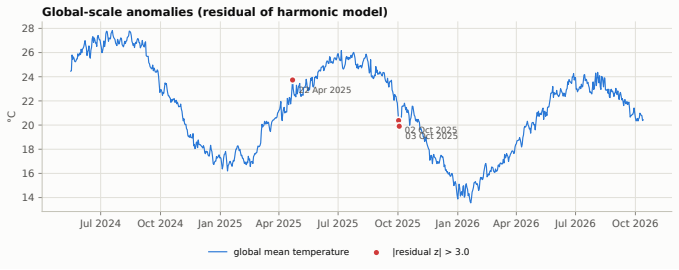


Figure 17. Remaining global-scale anomalies after cleaning ($|z| > 3$ on the residual of the harmonic model).

Table 2. Ten strongest Isolation-Forest anomalies.

Date	Location	Country	Temp (°C)	Wind (kph)	Precip (mm)	PM2.5
2026-07-11	Tarawa	Kiribati	24.6	8.3	6.3	3.1
2024-07-03	Bridgetown	Barbados	25.2	33.1	4.6	1.9
2025-07-08	Islamabad	Pakistan	26.3	5.4	6.3	85.3
2024-05-27	Dhaka	Bangladesh	25.3	45.0	6.3	19.3
2025-05-29	Dhaka	Bangladesh	24.8	45.0	6.3	15.9
2025-02-27	Islamabad	Pakistan	13.9	5.8	6.3	87.1
2026-03-01	Dodoma	Tanzania	18.1	3.6	6.3	6.3
2024-08-23	Asmara	Eritrea	14.2	15.1	3.7	1.3
2025-08-24	Khartoum	Sudan	24.1	4.7	3.7	29.0
2024-06-16	Guatemala City	Guatemala	16.9	11.2	6.1	183.9

Every model is a nested MLflow run with its parameters, validation and test metrics, a residual plot, the model artefact and an input signature. Each one is also registered in the MLflow Model Registry (`weather-temp-<model>`).

Metrics. MAE, RMSE, sMAPE, R^2, and the share of forecasts within 2 °C. MAPE is computed only where $|y| \geq 1$ °C, because it explodes near 0 °C. We also report *skill*, the RMSE reduction relative to persistence (tomorrow = today).

5.1. City panel: single models and ensembles

Four learners are compared against persistence:

- ridge regression on the train-fitted standardisation;
- random forest;
- XGBoost and LightGBM, both with early stopping on the validation window.

Three ensembles are built from validation-window predictions only:

- a simple mean;
- an inverse-validation-RMSE weighted mean;
- **stacking** with non-negative least squares. Its weights are XGBoost 0.49, LightGBM 0.26, Ridge 0.24 and random forest 0.01.

The stacked ensemble is logged as an MLflow `pyfunc` model (`weather-temp-ensemble`). The pipeline checks that the served model reproduces the offline predictions.

All learned models beat persistence. The ensembles beat every single model, and stacking is best: test RMSE 1.838 °C, MAE 1.21 °C, $R^2 = 0.929$, 81 % of forecasts within 2 °C, and 12.2 % skill. Ridge receives a meaningful weight despite being the weakest member because its errors are the least correlated with the tree models'. The

Table 3. Next-day temperature forecast accuracy on the 60-day test window (all cities). Skill = RMSE reduction relative to persistence.

	MAE	RMSE	sMAPE (%)	R^2	≤2 °C (%)	Skill (%)	Val RMSE
Persistence (baseline)	1.321	2.094	7.992	0.907	79.185	0.000	2.194
Ridge regression	1.310	1.916	7.678	0.922	79.934	8.527	1.990
Random forest	1.229	1.874	7.348	0.926	80.692	10.526	1.976
XGBoost	1.225	1.850	7.334	0.928	81.076	11.664	1.950
LightGBM	1.220	1.850	7.302	0.928	81.102	11.674	1.955
Ensemble: simple mean	1.218	1.844	7.263	0.928	81.352	11.971	1.944
Ensemble: inverse-RMSE weighted	1.218	1.844	7.263	0.928	81.325	11.976	1.944
Ensemble: stacking (NNLS)	1.211	1.838	7.233	0.929	81.361	12.247	1.940

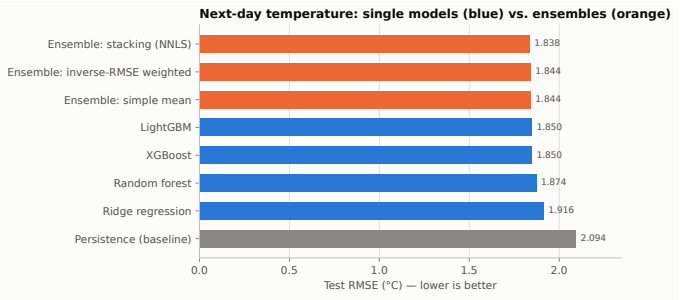


Figure 18. Test RMSE of single models (blue) and ensembles (orange).

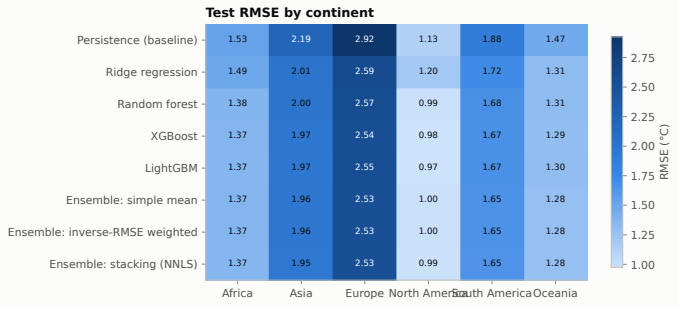


Figure 19. Test RMSE by continent.

gains look modest because day-to-day weather from a single daily snapshot is noisy; the hardest regions are mid-latitude Europe and Asia, where synoptic systems pass through (Figure 19).

5.2. Global mean series: multi-step trend forecasting

Each model is fitted on everything before the test window and forecasts all 60 test days in one shot:

- naive and seasonal-naive (value 365 days earlier) baselines;
- damped Holt–Winters with a 365-day season;
- harmonic regression (trend, four annual harmonics, observation-hour harmonics);
- regression with ARIMA(1,0,1) errors on the same design (`sarimax_fourier`);
- their mean ensemble.

The snapshot hour of future days is unknown, so it is held at its last 30-day average.

Table 4. Global mean temperature: 60-day multi-step forecast accuracy on the test window.

	mae	rmse	smape	r2
Naive	2.077	2.275	9.166	-5.002
Seasonal Naive	1.412	1.522	6.187	-1.685
Holt Winters	0.402	0.517	1.899	0.690
Harmonic Regression	0.275	0.355	1.275	0.854
Sarimax Fourier	0.292	0.368	1.344	0.843
Ensemble Mean	0.286	0.346	1.330	0.861

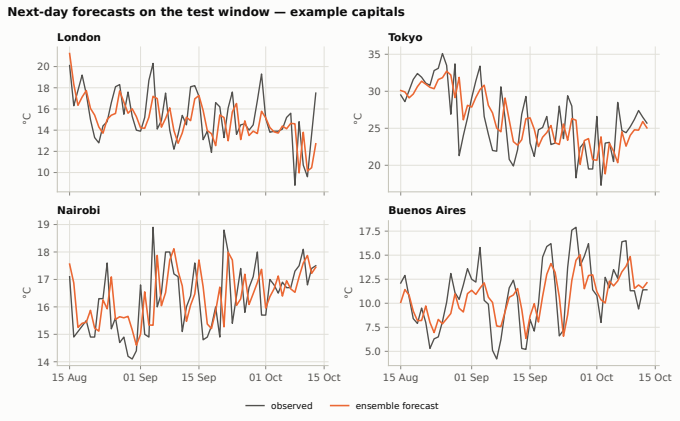


Figure 20. Stacked-ensemble next-day forecasts against observations for four example capitals.

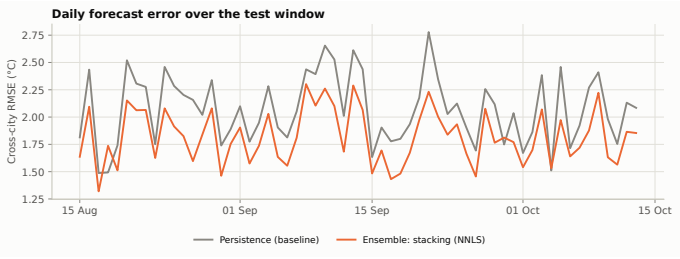


Figure 21. Cross-city RMSE per day over the test window.

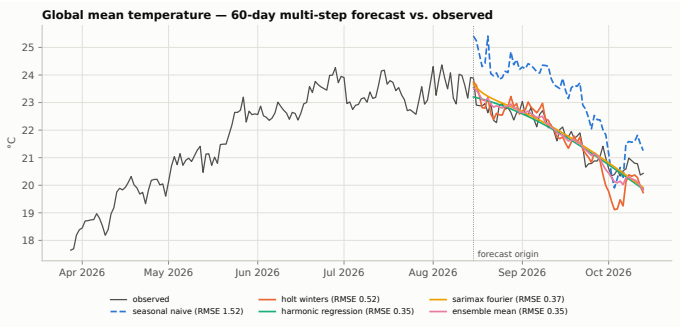


Figure 22. 60-day multi-step forecasts of the global mean temperature.

Harmonic regression and SARIMAX reach about 0.35 °C RMSE over a 60-day horizon, four times better than seasonal-naive. The ensemble is marginally best. Including the observation-hour terms cuts the SARIMAX error from 0.62 to 0.37 °C, which confirms that the confounder matters for forecasting as well as for trend estimation. Refitted on all data, the best interval-producing model projects the next 90 days with a 95 % prediction interval (Figure 23).

6. Advanced analyses

6.1. Climate analysis

For every city we fit temperature on a linear trend, two annual harmonics, and two diurnal harmonics of the snapshot hour. The record covers only about 2.4 years, so these slopes describe short-term variation, not climate change. Without the hour terms, 79 % of cities appear to cool, at a median of −1.79 °C/yr. With them the median slope falls to −0.51 °C/yr (Figure 24). The diurnal effect captured by the hour terms is large: about 8 °C peak-to-peak in the median city. Seasonality is structured by latitude. The two hemispheres run in opposite phase (Figure 25), and the annual range grows strongly with |latitude| ($r = 0.84$). Continental capitals such as Ulaanbaatar, Astana and Bishkek have the widest swings (30–42 °C) (Figure 26).

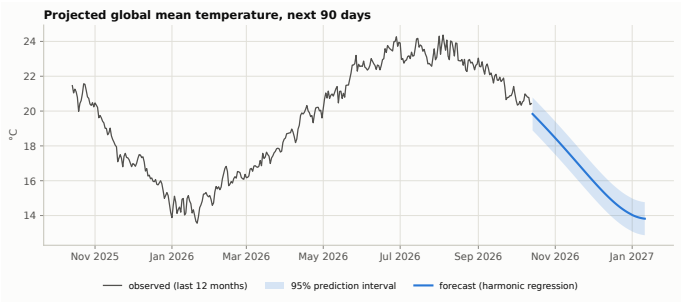


Figure 23. Projected global mean temperature for the next 90 days.

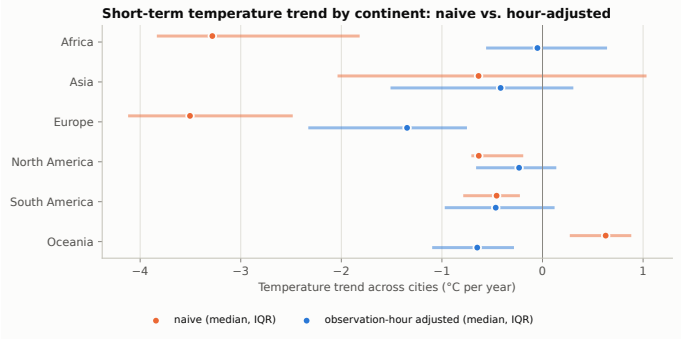


Figure 24. Per-city temperature trends by continent, naive and observation-hour adjusted.

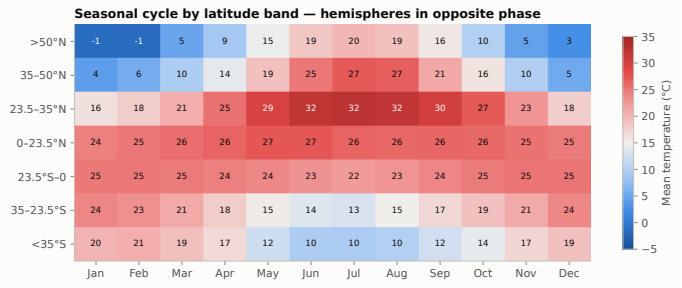


Figure 25. Mean temperature by latitude band and calendar month.

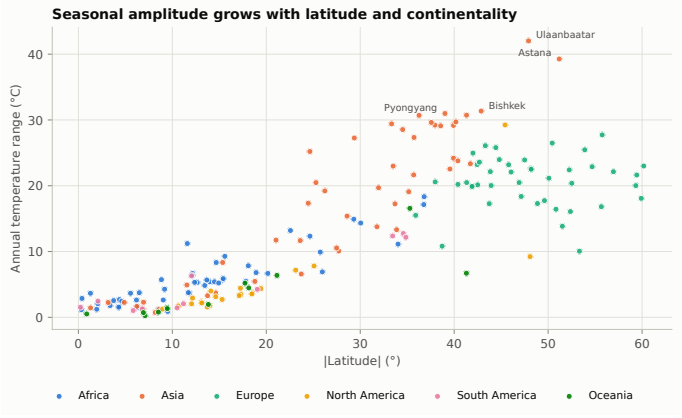


Figure 26. Annual temperature range against |latitude|.

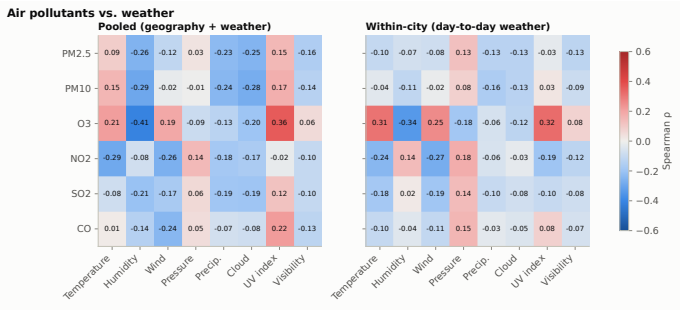


Figure 27. Air pollutants against weather: pooled (left) and within-city (right) Spearman correlations.

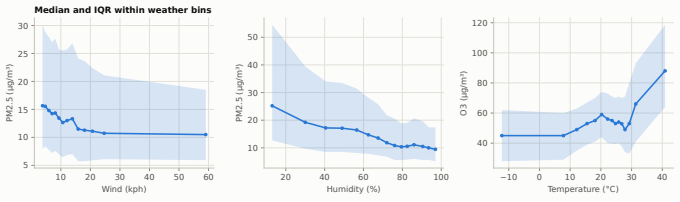


Figure 28. Pollutant response within weather bins (median and IQR).

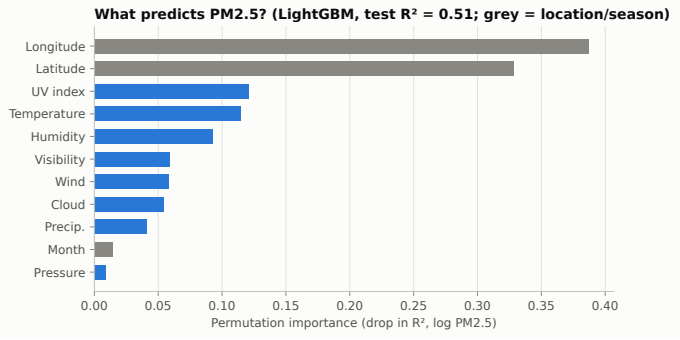


Figure 29. Air-quality drivers and hotspots. — Drivers of PM2.5.

6.2. Environmental impact: air quality and weather

Pooled correlations mix geography with weather: humid tropical capitals are cleaner than dry, dense ones. To isolate the weather effect we also compute *within-city* correlations, after removing each city's mean (Figure 27).

- **Ozone** is photochemical. It rises with temperature ($\rho = 0.31$) and UV ($\rho = 0.32$) and falls with humidity.
- **NO_2** and **SO_2** fall with wind (dispersion) and rise with pressure (stagnant anticyclones).
- **PM2.5** falls with wind, precipitation and cloud (wash-out and mixing). Within a city these effects are weak ($|\rho| \leq 0.13$), so particulate levels are driven mostly by local emissions.

A LightGBM model of log PM2.5 reaches test $R^2 \approx 0.51$. Location (longitude and latitude) dominates its permutation importance, calendar month barely matters, and UV, temperature and humidity are the main weather drivers (Figure 29). The median PM2.5 of 41 % of capitals exceeds the WHO 24-hour guideline of 15 µg/m³. Asia carries the heaviest burden.

6.3. Feature importance

Feature importance for the LightGBM next-day model is assessed four ways: split **gain**, **permutation** (increase in test RMSE), **mean |SHAP|** (TreeExplainer) and **mutual information**. All four agree that persistence dominates: current temperature first, then its 3-, 7- and 14-day rolling means and the feels-like temperature (Table 5). Gain and SHAP rankings correlate at $\rho = 0.73$, and permutation

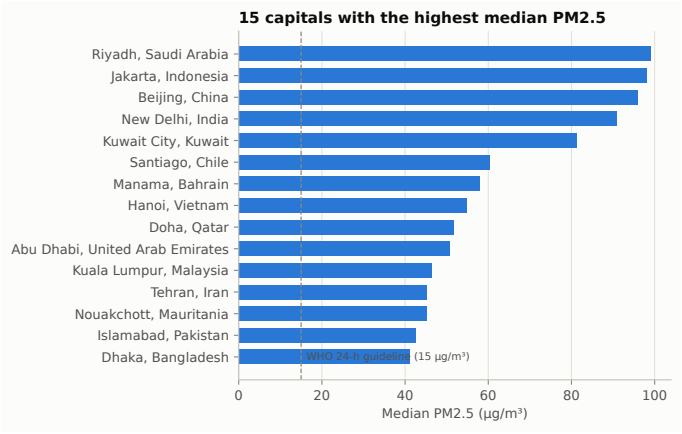


Figure 30. Air-quality drivers and hotspots. — Most polluted capitals.

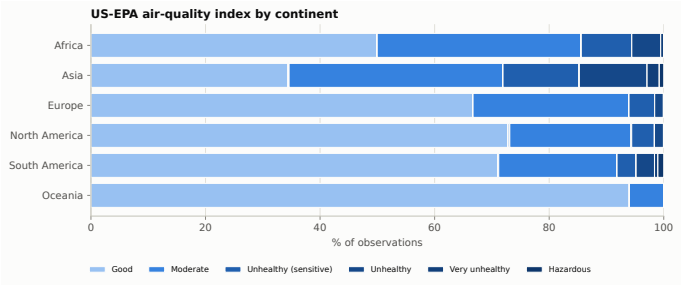


Figure 31. US-EPA air-quality index categories by continent.

Table 5. Top-15 features for next-day temperature under four importance techniques (sorted by mean rank).

	LightGBM gain	Permutation (ΔRMSE)	Mean |SHAP|	Mutual info.	Mean rank
temp_t	0.593	3.086	3.014	1.432	1.000
temp_rollmean3	0.160	0.372	0.898	1.357	2.250
temp_rollmean14	0.028	0.475	0.716	1.253	3.500
temp_rollmean7	0.057	0.070	0.325	1.296	4.250
feels_like_celsius	0.135	0.180	0.693	1.198	4.250
temp_lag14	0.002	0.023	0.191	0.929	8.250
latitude	0.001	0.064	0.142	0.733	9.250
humidity	0.001	0.042	0.206	0.441	10.000
pressure_mb	0.002	0.039	0.129	0.246	10.750
longitude	0.001	0.015	0.082	0.655	14.750
temp_diff1	0.002	0.016	0.083	0.051	16.000
pressure_diff1	0.001	0.015	0.060	0.129	16.000
uv_index	0.001	0.008	0.128	0.298	16.000
temp_lag1	0.001	0.010	0.028	1.219	16.000
abs_latitude	0.001	0.021	0.081	0.713	16.750

and SHAP at $\rho = 0.83$. Mutual information also rewards redundant features, such as raw lags, that the trees do not need. Past the persistence block, the secondary signals are humidity, pressure and its daily change, latitude, UV, and the diurnal snapshot-hour terms (Figure 33).

6.4. Spatial analysis

Every variable is clustered in space. Moran's I with 8-nearest-neighbour great-circle weights and 999 permutations gives $p = 0.001$ for all four variables (Figure 35): temperature $I = 0.68$, humidity 0.71, precipitation 0.54 and PM2.5 0.31. PM2.5 is the least spatially structured because it depends on local emissions. Temperature declines away from the equator, faster in the Northern Hemisphere, where most high-latitude capitals lie (Figure 37). Interactive maps are produced in `reports/maps/`: a Folium map of all capitals with pop-ups, and Plotly maps.

6.5. Geographical patterns

Doha is the hottest capital in the sample (32.9 °C mean) and Ulaanbaatar the coldest (6.1 °C). Asia has both the widest temperature spread and the highest median PM2.5. Europe is cool and humid with clean air, while Africa and Oceania are warm

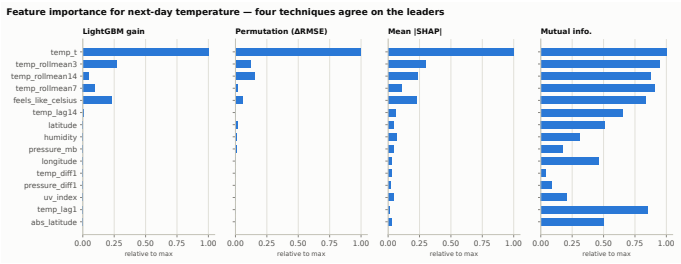


Figure 32. Top-15 features under four importance techniques, each scaled to its maximum.

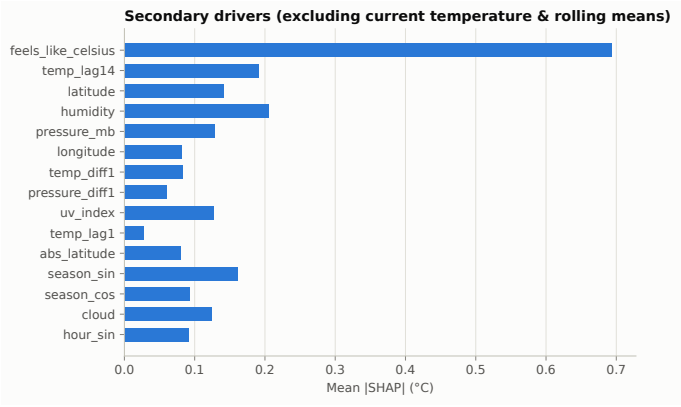


Figure 33. Feature importance beyond persistence, and a SHAP beeswarm. — Secondary drivers.

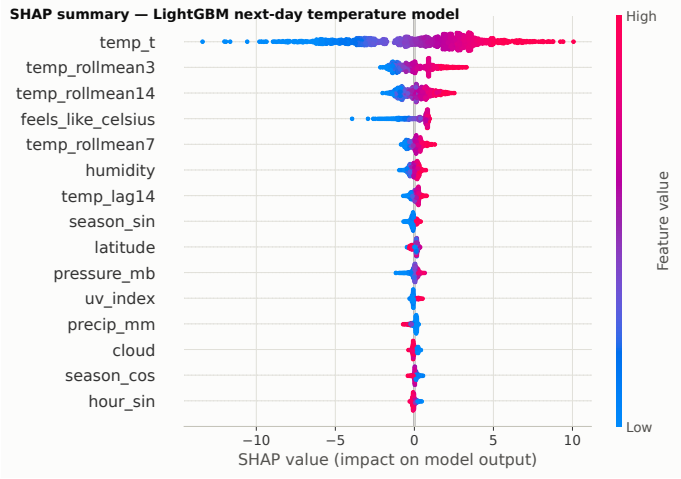


Figure 34. Feature importance beyond persistence, and a SHAP beeswarm. — SHAP summary.

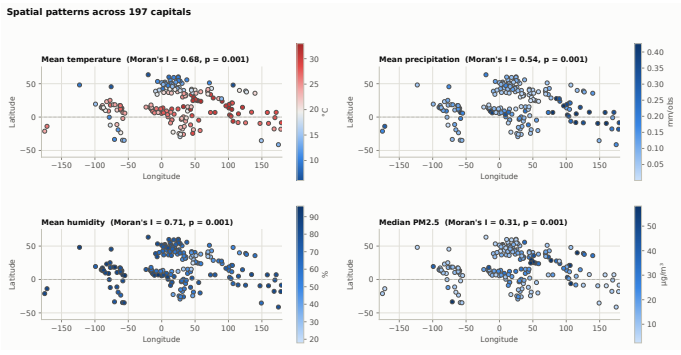


Figure 35. Spatial distribution of climate and air quality, with Moran's I.

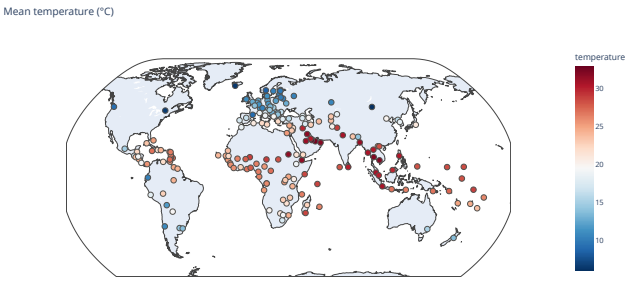


Figure 36. Mean temperature of the 197 capitals (natural-earth projection).

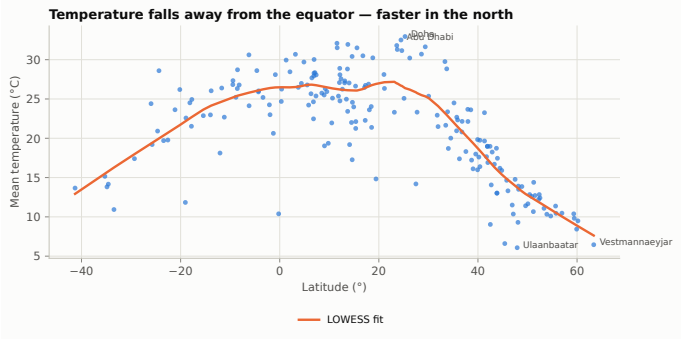


Figure 37. Latitude profile of mean temperature with a LOWESS fit.

Table 6. Weather and air-quality profile per continent.

continent	Cities	Temp (°C)	Temp SD	Precip (mm)	Humidity (%)	Wind (kph)	PM2.5 med.	UV
Africa	52	24.69	5.32	0.11	63.98	12.66	15.50	4.54
Asia	51	24.54	10.22	0.14	51.97	12.56	22.60	4.66
Europe	46	13.85	9.51	0.09	70.12	12.02	10.73	2.04
North America	23	22.44	6.59	0.12	84.14	13.59	9.60	0.85
South America	12	17.99	6.49	0.13	82.74	7.54	9.44	0.88
Oceania	13	24.75	5.58	0.27	80.63	18.38	5.55	0.42

with narrow ranges (Figure 38). Per-country summaries are in `reports/tables/country_summary.csv`.

7. Key insights

1. **Check how the data was collected before fitting any trend.** The local time of the daily snapshot drifts from afternoon to early morning, which creates an artificial cooling of about 1.8 °C/yr. Correcting for it removes most of that trend and improves multi-step forecasts by about 40 %.
2. **The source feed has failures of its own.** One whole day (17 Aug 2026) is corrupted across all cities. A batch-level detector catches it; a per-city outlier fence would not.
3. **Next-day temperature is mostly persistence.** The stacked ensemble improves RMSE on persistence by 12 % (1.84 against 2.09 °C), and ensembles beat every single model.
4. **Global seasonal trends are highly predictable.** The global mean is forecast 60 days ahead with an RMSE of about 0.35 °C.
5. **Air quality is mostly local.** Ozone follows sunshine and heat. PM2.5 is driven mostly by location, and weather explains only a small share of its day-to-day variation.
6. **Climate has strong spatial structure.** Temperature and humidity have Moran's I near 0.7, and the size of the annual cycle is largely set by latitude.

Limitations. The record covers about 2.4 years, so long-term climate trends cannot be estimated. There is one instantaneous snapshot per day, not a daily mean or extreme. Some locations are geocoding homonyms. Pollutant values come from a model-based feed.

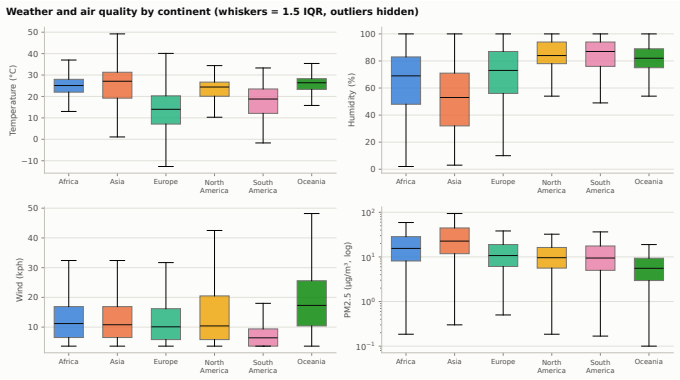


Figure 38. Weather and air quality by continent.

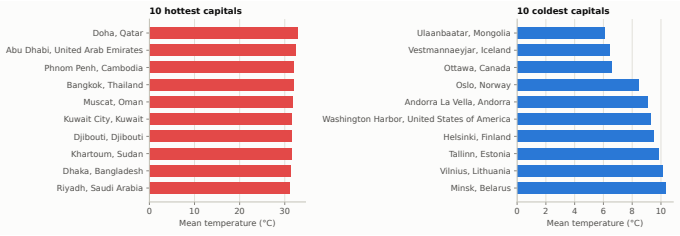


Figure 39. The hottest and coldest capitals by mean temperature.

Next steps.

- Aggregate the 15-minute API data, or re-weight observations by local hour.
- Add multi-horizon direct models for days 1–7.
- Add probabilistic forecasts (quantile LightGBM).
- Add ERA5 reanalysis as a long-term baseline.

8. Reproducibility

- **Environment**: `uv` (`pyproject.toml` and `uv.lock`), Python 3.12; formatting and linting with `black` and `ruff`, enforced by a pre-commit hook.
- **Configuration**: every setting (paths, MLflow, data source, cleaning thresholds, features, split, hyper-parameters) is in .env. It is loaded into a typed pydantic settings model with nested sections and validation, built once by a cached `get_settings()` (src/weather/utils/config.py). Environment variables override the file.
- **Pipeline**: `uv run dvc repro` rebuilds every artefact. Every stage depends on .env, and `dvc metrics diff` compares experiments.
- **Data versioning**: raw and processed data and model binaries are DVC-tracked and pushed to a DVC remote. Figures, tables and metrics are committed to git.
- **Experiment tracking**: `uv run mlflow ui -backend-store-uri sqlite:///mlflow.db` shows one run per stage, nested runs per model, artefacts, and registered models.
- **Tests**: `uv run pytest`. The suite covers cleaning rules, corrupted-batch detection, alignment of features and targets, strict time ordering of the split (no leakage), and the metrics.